

ADRD ePhenotyping with Fairness Analysis

Project Information

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Project Summary

This project evaluates a pre-trained Convolutional Neural Network (CNN) model for automated detection of Alzheimer's Disease and Related Dementias (ADRD) from clinical Electronic Health Record (EHR) notes, with comprehensive fairness analysis across demographic groups. The CNN model was originally developed by Knox and Obeid using MUSC clinical data.

Research Questions

1. Does the CNN model perform equitably across demographic groups (gender, race, ethnicity)?
 2. Do the discriminative linguistic features driving CNN predictions differ across demographic subgroups?
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Project Experiments

Aim 1: Demographic Fairness Analysis

Objective: Evaluate CNN classification parity across demographic subgroups using approximate randomization testing (10,000 permutations).

Subgroups Analyzed: Gender (2), Race (4), Ethnicity (2), Intersectional Gender x Race (4)

Aim 2: Feature-Level Fairness Analysis

Objective: Identify discriminative features and assess consistency across demographics using Chi-squared testing, TF-IDF analysis, and behavioral testing.

Features Tested: 13,889 clinical terms from 1,460 patient documents

Experimental Results

Dataset Characteristics

Characteristic	Value
Total Samples	1,460
ADRD Cases	657 (45.0%)
Control Cases	803 (55.0%)
Data Source	MUSC EHR Clinical Notes

Demographic Distribution

Demographic	Category	N	%
Gender	Female	828	56.7%
Gender	Male	632	43.3%
Race	White	1,013	69.4%
Race	Black	407	27.9%
Race	Asian	10	0.7%
Race	Other	21	1.4%
Ethnicity	Non-Hispanic	1,441	98.7%
Ethnicity	Hispanic	14	1.0%

Overall Model Performance (Best Model - Cycle 9)

Metric	Value	95% CI
AUC	0.9867	0.9818-0.9916
Accuracy	94.25%	-
Sensitivity	97.26%	-
Specificity	91.78%	-
Precision (PPV)	90.64%	-
NPV	97.62%	-
F1 Score	0.9383	-
Brier Score	0.0440	-

Confusion Matrix (n = 1,460)

	Predicted Control	Predicted ADRD
Actual Control	737 (TN)	66 (FP)
Actual ADRD	18 (FN)	639 (TP)

Aim 1 Results: Demographic Fairness

Gender Performance

Metric	Female	Male	Difference
AUC	0.9867	0.9875	0.0008
Sensitivity	98.40%	95.73%	2.67%
Specificity	90.71%	93.16%	2.45%

Finding: No statistically significant disparities ($p = 0.432$)

Racial Performance

Race	N	AUC	Sensitivity	Specificity
White	1,013	0.9855	97.04%	91.43%
Black	407	0.9893	97.83%	93.22%
Asian	10	1.0000	100%	100%
Other	21	0.9727	90.00%	81.82%

Finding: AUC difference (White-Black): 0.0038 (within fairness threshold)

Intersectional Performance (Gender x Race)

Intersection	N	AUC	Sensitivity	Specificity
Female x White	546	0.9839	98.23%	89.69%
Male x White	467	0.9880	95.56%	93.38%
Female x Black	260	0.9913	98.54%	92.68%
Male x Black	147	0.9853	96.77%	94.44%

Finding: AUC range: 0.0074 (within acceptable limits)

Aim 2 Results: Feature Analysis

Chi-Squared Test Results

Measure	Value
Features Tested	13,889
Significant Features ($FDR < 0.05$)	3,790
ADRD-Overrepresented	3,790
Control-Overrepresented	0

Top 10 Discriminative Terms for ADRD

Rank	Term	Chi-Squared	Clinical Category
1	goal	4,623	Care planning
2	outcome	4,401	Care planning
3	ongoing	3,717	Disease management
4	progressing	2,753	Disease progression
5	discharge	2,129	Care transitions

6	oral	2,114	Medication admin
7	pt	2,059	Patient reference
8	dementia	1,858	Diagnosis
9	admission	1,841	Hospital encounters
10	care	1,725	General care

Feature Consistency Across Demographics

Comparison	Overlap	Interpretation
Female vs. Male	90%	Highly Consistent
Black vs. White	70%	Good Consistency

Finding: 70-90% term overlap indicates robust, generalizable discriminative features

Key Findings

- 1. **Model Performance:** CNN achieves AUC = 0.9867 with 97.26% sensitivity for ADRD detection
 - 2. **Demographic Fairness:** No statistically significant performance disparities across gender, race, or ethnicity (all p-values > 0.05)
 - 3. **Intersectional Fairness:** AUC variance < 1% across Gender x Race intersections
 - 4. **Feature Generalizability:** 70-90% discriminative term overlap across demographic subgroups indicates model captures universal ADRD phenotypes
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Resources

Code Repository

- **GitHub:** https://github.com/gyasifred/adrd_ePheno

Datasets Used

- MUSC Electronic Health Records (de-identified clinical notes)
- Pre-trained CNN models (Knox/Obeid, 10 cycles)

Key Analysis Scripts

- 03_evaluate_models.R - Model evaluation
 - 04_demographic_analysis.R - Aim 1: Fairness analysis
 - 05_aim2_feature_analysis.R - Aim 2: Feature analysis
 - 06_integration_analysis.R - Integration analysis
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Acknowledgements

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Data Source: Medical University of South Carolina (MUSC) Research Data Warehouse
